



Functions in C++

❖ User define Function

- A User defined function in C++ is a function that is define by the programmer to perform a specific task and can be called whenever needed in the program.

- Syntax

```
void hira()  
{  
    // code to be executed  
}
```

Here,

hira() is the name of the function.

void means that the function does not have a return value.

inside the function (the body), **add code** that defines what the function should do.

Q1. WAP to add two number using function.

```
2.cpp X  
#include<iostream>  
#include<conio.h>  
using namespace std;  
void hira()  
{  
    int a,b;  
    cout << " Enter two number = " ;  
    cin >>a>>b;  
    cout << " sum of two number is = " <<a+b;  
}  
int main()  
{  
    cout << " Hello Hira" << endl;  
    hira();  
    getch();  
}
```

```
J:\c-c++\part-3\2.exe  
Hello Hira  
Enter two number = 8  
5  
sum of two number is = 13
```

Or (2nd method) → Use return value

```
#include<iostream>
#include<conio.h>
using namespace std;

int hira()
{
    int a,b;
    cout <<"Enter any two number : ";
    cin>>a>>b;
    return a+b;
}

int main()
{
    cout << " Welcome two number addition program" << endl;
    int a=hira();
    cout << "your sum is = " << a;
    getch();
}
```

```
C:\Users\hira\Desktop\hello.exe

Welcome two number addition program
Enter any two number : 20
50
your sum is = 70
```

❖ Function Execution Flow in C++

- When a function is called, the program's execution temporarily moves from the calling function to the called function, executes its statements, and then returns to the point where the function was called.

```
#include <iostream>
using namespace std;

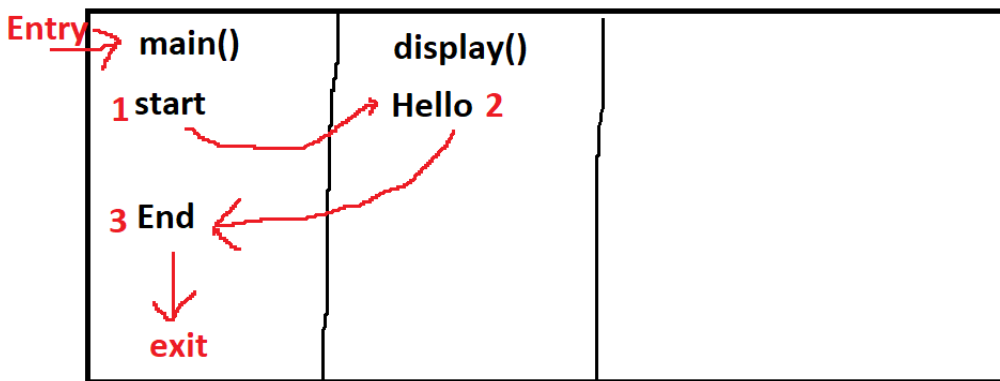
void display()
{
    cout << "Hello";
}

int main()
{
    cout << "Start" << endl;
    display();
    cout << "End" << endl;
    return 0;
}
```

```
C:\Users\hira\Desktop\hello.exe

Start
HelloEnd
```

RAM (2GB)



```
main()
↓
cout << "Start"
↓
display() ← Function call
↓
display() function starts
↓
cout << "Hello"
↓
display() ends
↓
Control returns to main()
↓
cout << "End"
↓
return 0
↓
Program ends
```

❖ Function Declaration

- A **function declaration** tells the compiler about the function name, return type, parameters and parameters data type before its actual definition.

- It is also called the **function prototype**.
- The process of written function prototype/declaration is called **function prototyping**.

❖ Function Definition

- A **function definition** contains the actual code that specifies what the function does.

❖ Function call

- To **call a function**, write the function's name followed by two parentheses () and a semicolon ;

```
#include<iostream>
#include<conio.h>
using namespace std;
void hira();
int main()
{
    cout << " Hello Hira" << endl;
    hira();
    getch();
}
void hira()
{
    int a,b;
    cout << " Enter two number = " ;
    cin >>a>>b;
    cout << " sum of two number is = " <<a+b;
```

Diagram illustrating function call and definition in the provided code:

- `void hira();` is labeled as **function declaration**.
- `hira();` inside the `main` function is labeled as **call the function**.
- `void hira() { ... }` is labeled as **function definition**.

❖ Difference b/w fun declaration & fun definition

Function Declaration	Function Definition
Tells the compiler that a function exists	Tells the compiler what the function does
Also called a function prototype	Contains the actual implementation
Does not contain the function body	Contains the function body { }

Ends with ;

Does not end with ; after the function body

- ✓ **Note:** If a user-defined function, such as **hira()**, is defined after the **main()** function without a prior declaration, an error will occur when **hira()** is called inside **main()**. This is because the compiler processes the program from top to bottom and must know the function's declaration before it is used.

```
1  #include<iostream>
2  #include<conio.h>
3  using namespace std;
4  int main()
5  {
6      cout << " Hello Hira" << endl;
7      hira();
8      getch();
9  }
10 void hira()
11 {
12     int a,b;
13     cout << " Enter two number = " ;
14     cin >>a>>b;
15     cout << " sum of two number is  = " <<a+b;
16 }
17
```

error (Reason :- Not function declaration)

❖ Retrun in c++

- The return statement in C++ is used to **send a value back** from a function to the place where the function was called.
- return **terminates the current function**.
- It can **send a value back** to the calling function.
- A non-void function generally returns a value of its declared type.
- A void function can use return; without a value.

(2nd method) → Use return value

```
#include<iostream>
#include<conio.h>
using namespace std;

int hira()
{
    int a,b;
    cout <<"Enter any two number : ";
    cin>>a>>b;
    return a+b;
}

int main()
{
    cout << " Welcome two number addition program" << endl;
    int a=hira();
    cout << "your sum is = " << a;
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
Welcome two number addition program
Enter any two number : 20
50
your sum is = 70
```

- A void function can use return; without a value.

```
#include <iostream>
#include<conio.h>
using namespace std;

void hira(int x)
{
    cout << " Value of x is = " << x ;
    return ;
}

int main()
{
    int a=10;
    hira(a); // function call
    getch();
    return 0;
}
```

C:\Users\hira\Desktop\hello.exe

```
Value of x is = 10
```

```
1  #include <iostream>
2  #include<conio.h>
3  using namespace std;
4
5  void hira(int x)
6  {
7
8      cout << " Value of x is = " << x ;
9      return 0 ;
10 }
11
12 int main()
13 {
14     int a=10;
15     hira(a); // function call
16     getch();
17     return 0;
18 }
19
20
```

➤ Returning One Value

- When a user-defined function returns only one value to the calling function using the return statement, it is called returning one value.

- Syntax :

```
return_type function_name(parameters)
{
    // statements
    return value;
}
```

➤ Returning Multiple Values

- In C++, a normal function can directly return only one value using the return statement. But we can return multiple values indirectly using references, pointers, arrays, structures/classes, or pair.

Q. Write a program to print sum and multiplication any two number using user define function and print final answer in main() function.

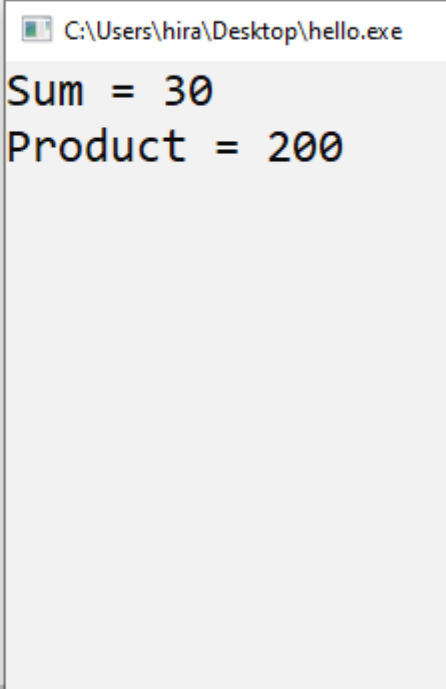
```
#include <iostream>
#include<conio.h>
using namespace std;

void Hira(int a, int b, int &sum, int &product)
{
    sum = a + b;
    product = a * b;
}

int main()
{
    int s, p;

    Hira(10, 20, s, p);

    cout << "Sum = " << s << endl;
    cout << "Product = " << p;
    getch();
    return 0;
}
```



❖ Global and Local variables

The variable declared in a program may broadly be classified into following two types;

- (i) Local variable
- (ii) Global variable

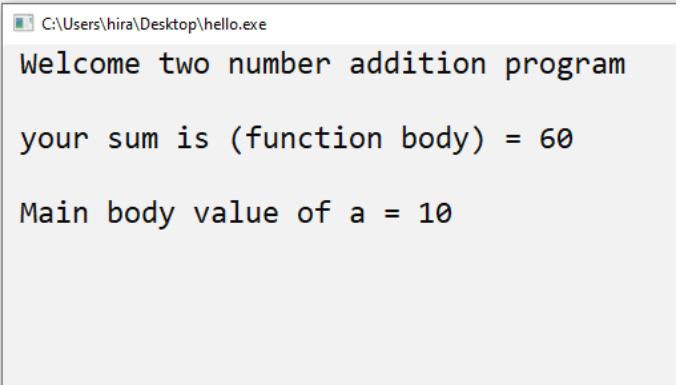
(i) Local Variable

- Local variable A variable declared within the body of a function will be evaluated only within the function.

```
#include<conio.h>
using namespace std;

void hira()
{
    int a=20,b=40; //local variable
    cout << " your sum is (function body) = " << a+b << endl << endl;
}

int main()
{
    int a=10; // local variable
    cout << " Welcome two number addition program" << endl << endl;;
    hira();
    cout << " Main body value of a = " << a;
    getch();
}
```



(ii) Global variable

- A variable that is declared outside any function is known as a global variable. These variables are available to all functions which follow their declaration.
- If a is variable then $\rightarrow ::a$ represents the global variable. The symbol $::$ is called scope resolution operator.
- It is also known as External variables

```
#include<iostream>
#include<conio.h>
using namespace std;
int a=99; // global variable
void hira()
{
    int a=20,b=40; //local variable
    cout << " your sum is (function body) = " << a+b << endl;
    cout << " your sum is (function body) = " << ::a+b << endl << endl;
}
int main()
{
    int a=10; // local variable
    cout << " Welcome two number addition program" << endl << endl;;
    hira();
    cout << " (Main body) local value of a = " << a<<endl;
    cout << " (Main body) global value of a = " << ::a<<endl;
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

Welcome two number addition program

your sum is (function body) = 60

your sum is (function body) = 139

(Main body) local value of a = 10

(Main body) global value of a = 99

Questions

1. WAP to find sum of digit of a number. (input: 1234 $\rightarrow$ output: 10)
Use (i) function definition
(ii) function call
2. WAP to Print ABCD,.....Z series. (output $\rightarrow$ A B C D E F ,.....,X Y Z).
Use (i) Function declaration
(ii) function call
(iii) function definition
3. Write a C++ program to accept a character from the user and display its ASCII value. (input : @ $\rightarrow$ output: 64) (ASCII value range 0-127)
Use (i) Function declaration
(ii) function call
(iii) function definition
(iv) output statement write in main body i.e. main function.
4. Write a C++ program to perform addition, subtraction, multiplication, and division using separate user-defined functions. Accept the choice and numbers from the user, pass them to the appropriate function, and display the result in the main() function.

5. Write a C++ program to convert a given number of days into years, months, and days using user-defined functions (if year fix 365 days and month fix 30 days). {input: 32 → output: 0 year 1 month 2 days}

❖ Parameters

- Parameters act as variables inside the function.
- Every parameter contain datatype and variable name.
 - ✓ Syntax : datatype function_name(parameter1, parameter2,.....)
 - ✓ Example : hira(int a); or hira(int a, int b),....
- Information can be passed to functions as a parameter.
- Parameters are specified after the function name, inside the parentheses.
- You can add as many parameters as you want, just separate them with a comma.
- It is also called **formal parameters**.

❖ Argument

- An argument is the actual value, variable, or expression passed to a function when the function is called.
- It is also called **actual parameters**.

❖ Actual Parameters & Formal Parameters

In C++, Actual parameters is arguments generally refer to the values passed during a function call, while Formal parameters are the variables declared in the function definition.

Short :

Parameter = Used when defining a function.

Argument = Used when calling a function.

Formal parameters = Function definition

Actual parameters = Function call

```
#include<iostream>
#include<conio.h>
using namespace std;

void hira(int a, int b) // formal parameters
{
    cout << " sum of a + b = " << a+b << endl;
}

int main()
{
    cout << " ===== Welcome ===== " << endl;
    int a=10, b=40;
    hira(a,b); // argument (or) actual parameters
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
===== Welcome =====
sum of a + b = 50
```

OR

```
#include<iostream>
#include<conio.h>
using namespace std;

void hira(int a, int b) // formal parameters
{
    cout << " sum of a + b = " << a+b << endl;
}

int main()
{
    cout << " ===== Welcome ===== " << endl;
    hira(10,40); // argument (or) actual parameters
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
===== Welcome =====
sum of a + b = 50
```

❖ Types of parameters

(i) Default Parameter Value

You can also use a default parameter value, by using the equals sign (=). If we call the function without an argument, it uses the default value (10):

```
#include<iostream>
#include<conio.h>
using namespace std;

void hira(int a=10) // formal parameters
{
    cout << " value of a is = " << a << endl;
}

int main()
{
    cout << " ===== Welcome ===== " << endl;
    hira(20); // argument (or) actual parameters
    hira(199);
    hira();
    hira(255);
    getch();
}
```

C:\Users\hira\Desktop\hello.exe

```
===== Welcome =====
value of a is = 20
value of a is = 199
value of a is = 10
value of a is = 255
```

Here main() function

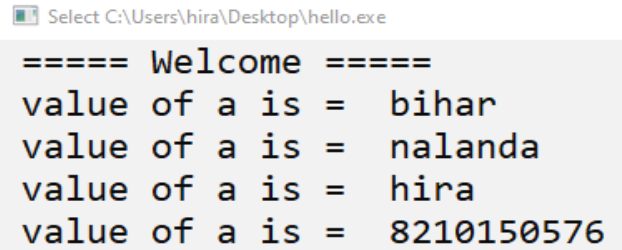
hira(); no any value pass in formal parameters, so here, hira() is called **default arguments**.

OR

```
#include<iostream>
#include<conio.h>
using namespace std;

void hira(string a="hira") // formal parameters
{
    cout << " value of a is = " << a << endl;
}

int main()
{
    cout << " ===== Welcome ===== " << endl;
    hira("bihar"); // argument (or) actual parameters
    hira("nalanda");
    hira();
    hira("8210150576");
    getch();
}
```



```
===== Welcome =====
value of a is =  bihar
value of a is =  nalanda
value of a is =  hira
value of a is =  8210150576
```

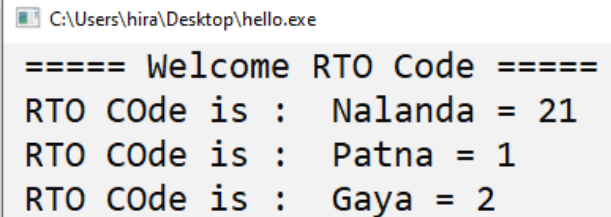
(ii) Multiple Parameters value

Inside the function, you can add as many parameters as you want:

```
#include<iostream>
#include<conio.h>
using namespace std;

void hira(string a, int b) // formal parameters
{
    cout << " RTO COde is :  " << a << " = " << b << endl;
}

int main()
{
    cout << " ===== Welcome RTO Code ===== " << endl;
    hira("Nalanda", 21); // argument (or) actual parameters
    hira("Patna", 01);
    hira("Gaya", 02);
    getch();
}
```



```
===== Welcome RTO Code =====
RTO COde is :  Nalanda = 21
RTO COde is :  Patna = 1
RTO COde is :  Gaya = 2
```

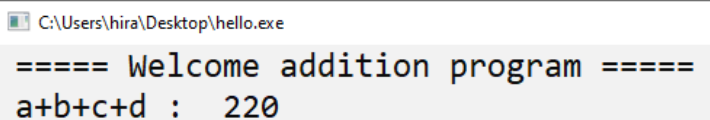
OR

```
#include<iostream>
#include<conio.h>
using namespace std;

void hira(int a, int b, int c, int d) // formal parameters
{
    cout << " a+b+c+d :  " << a+b+c+d << endl;
}

int main()
{
    cout << " ===== Welcome addition program ===== " << endl;
    int a=100, b=50;
    hira(a,b,20,50); // argument (or) actual parameters

    getch();
}
```



```
===== Welcome addition program =====
a+b+c+d :  220
```

Question :

6. WAP to print any 5 state name, capital and RTO code.
7. WAP to add three number. User input take in main function and print add() function.

8. Write a C++ program to create a user-defined function with your own name. The function should calculate and return the square of a given number.
9. Write a C++ program to create a user-defined function with your own name. The function should accept two numbers and return the maximum value.
10. Write a C++ program to create a user-defined function with your own name. The function should accept two numbers and return the minimum value.
11. Write a C++ program to create a user-defined function with your own name. The function should calculate and return the factorial of a given number.
12. Write a C++ program to create a user-defined function with your own name. The function should calculate and return the prime of a given number.
13. Write a C++ program to create a user-defined function with your own name. The function should calculate and return the leap year of a given number.
14. Write a C++ program to create a user-defined function with your own name. The function should calculate and return the armstrong of a given number.
15. Write a C++ program to create a user-defined function with your own name. The function should accept two numbers and return the both number find odd or even.

❖ Type of function based on arguments and return type

Sl. NO.	Type name	Argument	Return
1.	Take Nothing Return Nothing (TNRN) :- No argument, No return value	✗	✗
2.	Take Something Return Nothing (TSRN)	✓	✗

	:- Pass argument, No return value		
3.	Take Nothing Return Something (TNRS) :- No argument, with return value	✗	✓
4.	Take something Return something (TSRS) :- Argument, with return value	✓	✓

1. Take Nothing Return Nothing (TNRN)

```
void show()
{
    cout << "Hello";
}
```

```
#include <iostream>
#include <conio.h>
using namespace std;

void hira()
{
    int a,b;
    cout << " Enter two number :";
    cin >> a >> b;
    cout << " SUM = " << a+b;
    // return nothing
}

int main()
{
    hira(); // take nothing
    getch();
    return 0;
}
```

C:\Users\hira\Desktop\hello.exe

```
Enter two number :10
20
SUM = 30
```

2. Take Something Return Nothing (TSRN)

```
void show(int a)
{
    cout << a;
}
```

```

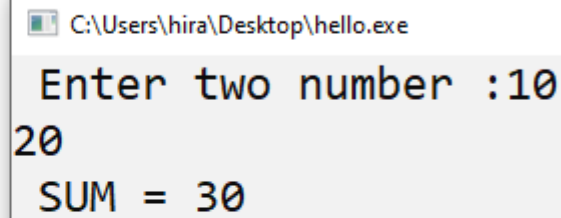
#include <iostream>
#include<conio.h>
using namespace std;

void hira(int x, int y)
{

    cout << " SUM = " << x+y;
    // return nothing
}

int main()
{
    int a,b;
    cout << " Enter two number :";
    cin >> a >> b;
    hira(a,b); // take something
    getch();
    return 0;
}

```



```

C:\Users\hira\Desktop\hello.exe
Enter two number :10
20
SUM = 30

```

3. Take Nothing Return Something (TNRS)

```

int show()
{
    return 10;
}

```

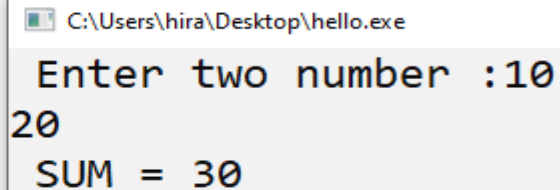
```

#include <iostream>
#include<conio.h>
using namespace std;

int hira()
{
    int a,b;
    cout << " Enter two number :";
    cin >> a >> b;
    return a+b; // return something
}

int main()
{
    int x;
    x = hira(); // take nothing
    cout << " SUM = " << x;
    getch();
    return 0;
}

```



```

C:\Users\hira\Desktop\hello.exe
Enter two number :10
20
SUM = 30

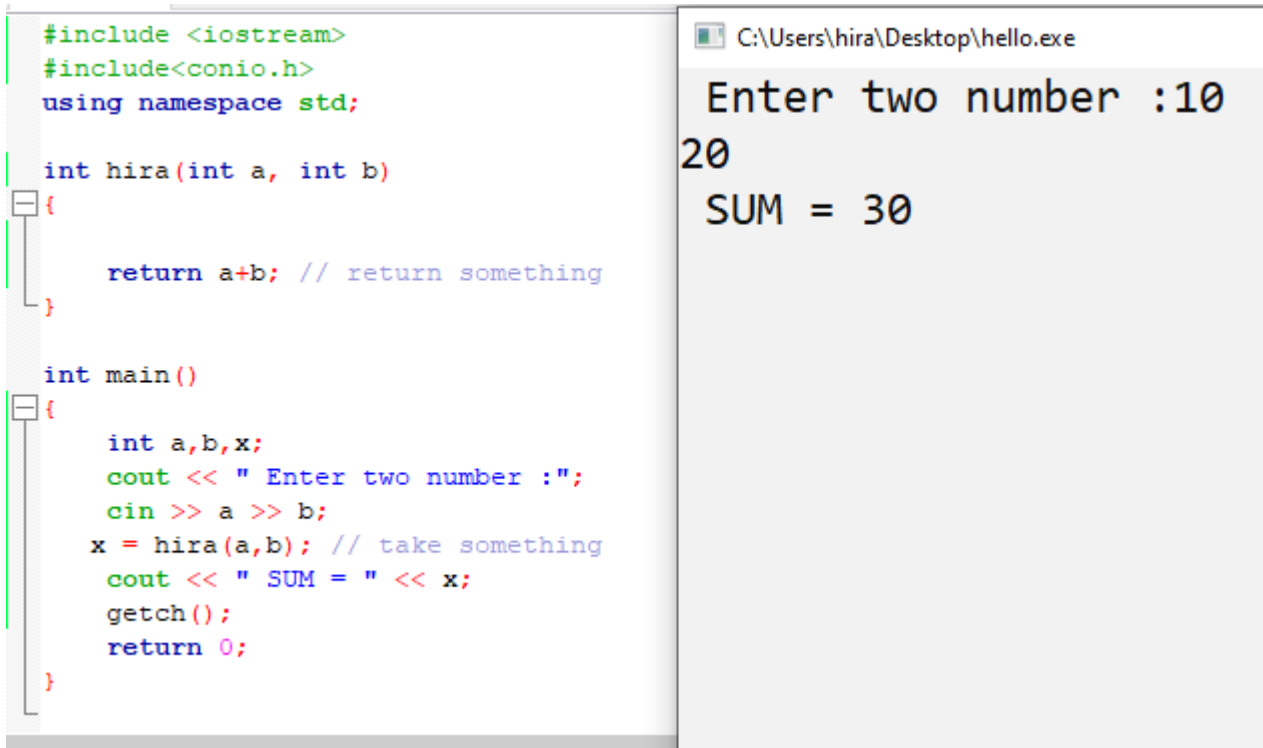
```

4. Take something Return something (TSRS)

```

int add(int a, int b)
{
    return a + b;
}

```



The image shows a C++ program in a code editor on the left and its execution output in a console window on the right. The code defines a function `hira` that takes two integers and returns their sum, and a `main` function that prompts the user for two numbers, calls `hira`, and prints the result.

```
#include <iostream>
#include<conio.h>
using namespace std;

int hira(int a, int b)
{
    return a+b; // return something
}

int main()
{
    int a,b,x;
    cout << " Enter two number :";
    cin >> a >> b;
    x = hira(a,b); // take something
    cout << " SUM = " << x;
    getch();
    return 0;
}
```

The console window shows the program's execution: it prompts for two numbers, the user enters 10 and 20, and the program outputs "SUM = 30".

Q. Write C++ programs to find whether a number is odd or even using all types of functions based on arguments and return values.

Q. Write C++ programs to print your name using all types of functions based on arguments and return values.

=====HIRA KUMAR=====